



Software Requirements Specification:

TOBLERONE - Restaurant Management & Customer Insights System

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Chapter 1 Introduction

This document describes the Software Requirements Specification (SRS) for the **Restaurant Discovery & Review Platform**, a two-sided web application developed as a training project.

1.1 Purpose

The purpose of this document is to describe the technology stack and functional requirements of the Restaurant Discovery & Review Platform. **Restaurant Owners** can register their restaurants and manage menus, while **Customers** can discover restaurants, read and write reviews, and track their personal dining expenses. The project is built end-to-end under the working names `my-phoenix` (frontend) and `my-toblerone` (backend).

1.2 Scope

This project covers the design and development of a restaurant platform with two types of users: restaurant owners and customers. Restaurant owners can create and manage their restaurant profile and menu through a dedicated dashboard. Customers can browse and search restaurants, view menus, submit reviews and ratings, and log their dining expenses through a personal budget tracker.

The system will be developed using the following components:

- A web-based frontend (`my-phoenix`) built with React, Next.js, TypeScript, and Ant Design
- A REST API backend (`my-toblerone`) built with Laravel
- A PostgreSQL database for storing all application data

The scope of this project does not include:

- Real payment gateway integration
- Geo-based ("near me") location search
- In-app table ordering
- Multi-currency support

Chapter 2 Technology Stack

2.1 Frontend – my-phoenix

- React 19.x
- Next.js 16.x (App Router)
- TypeScript 5.x
- Ant Design (AntD) 5.x
- TanStack React Query 5.x
- Axios
- Atomic Design structure: atoms/, icons/, molecules/, organisms/, templates/
- Deployment: Vercel

2.2 Backend – my-toblerone

- Laravel (PHP 8.2)
- Laravel Sail (Docker development environment)
- Laravel Sanctum (authentication)
- Laravel Scout (search abstraction)

2.3 Database / Infrastructure

- PostgreSQL (primary relational database)
- Redis (caching, sessions, queue driver)
- Typesense (search engine, via Laravel Scout)
- MinIO / RustFS (S3-compatible object storage)
- Mailpit (local email testing)

2.4 Dev / Debug Tools

- Laravel Tinker (REPL)

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- Laravel Telescope (request/query/exception inspector)
 - Laravel Horizon (queue monitoring)
 - Scribe (API documentation)
 - Postman (API testing)

Chapter 3 Functional Requirements

3.1 FR-1: Authentication & Authorization

- Register as Owner or Customer
- Login/logout using Laravel Sanctum
- Role-based access to protected endpoints

3.2 FR-2: Restaurant Management (Owner)

- Create, update, and delete restaurant profile
- Upload logo and cover image
- View list of owned restaurants

3.3 FR-3: Menu Management (Owner)

- Create and manage menu categories
- Add, edit, delete menu items (name, price, image, availability)

3.4 FR-4: Restaurant Discovery (Customer)

- Browse paginated restaurant listings
- Search restaurants by name, cuisine, or dish
- Filter by price range and rating
- View restaurant profile and full menu

3.5 FR-5: Reviews & Ratings

- Submit rating and written review
- Edit or delete own review
- Average rating calculated per restaurant
- Owner can reply to reviews

- Owner notified on new review

3.6 FR-6: Dining Budget Tracker (Customer)

- Log dining expense entries (restaurant, amount, date, notes)
- View monthly dining expense summary
- Receive monthly summary email

3.7 FR-7: Search Infrastructure

- Restaurant and menu data indexed via Typesense/Scout
- Typo-tolerant, partial-match search